https://www.math.ucdavis.edu/~mgaerlan/

$(x^2+y^2-1)^2$

Critical points: $\{(0,0)\}\cup\{(x,y)\in\mathbb{R}^2 \mid \underbrace{x^2+y^2=1}\}$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 48(x^2+y^2)^2 - 64(x^2+y^2)+16$$
$$= 48(1)^2 - 64(1) + 16$$
$$= 0 \Rightarrow \text{inconclusive}$$

$$\frac{dx_1}{dt} = ax_1 + bx_2$$

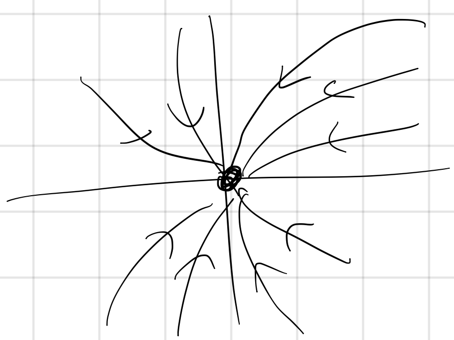
$$\frac{dx_2}{dt} = cx_1 + dx_2$$

$$\Longrightarrow \begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{d\vec{x}}{dt} = A \cdot \vec{x}$$

$$\vec{x}(t) = \begin{bmatrix} f(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix}$$

$$\left.\begin{aligned} ax_1 + bx_2 &= 0 \\ cx_1 + dx_2 &= 0 \end{aligned}\right\} \text{fixed point @ } (0,0)$$



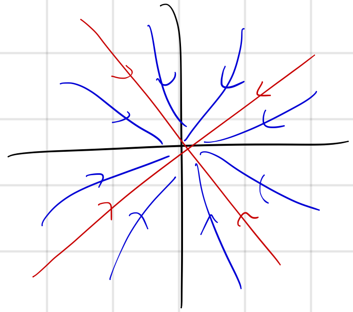
$$\vec{x}(t) = c_1 e^{\lambda_1 t}\vec{v}_1 + c_2 e^{\lambda_2 t}\vec{v}_2$$

λ_1, λ_2 are eigenvalues of A

$\vec{v}_1, \vec{v}_2$ are eigenvectors

eigenvalues real

$\tau^2 - 4\Delta > 0$

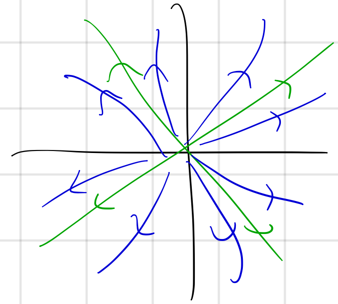


stable node

$\Delta > 0$

$\tau < 0$

eigenvalues same sign and negative

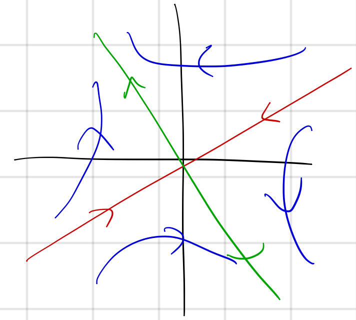


unstable node

$\Delta > 0$

$\tau > 0$

eigenvalues same sign and positive



saddle node

$\Delta < 0$

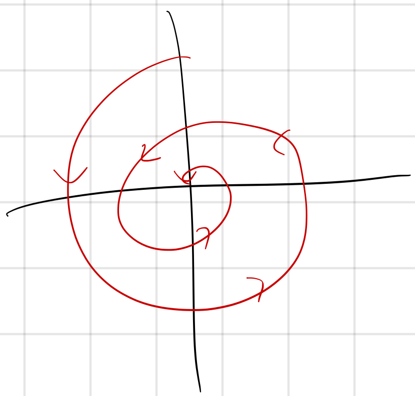
eigenvalues are different sign

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

eigenvalues Imaginary

$\tau^2 - 4\Delta < 0$

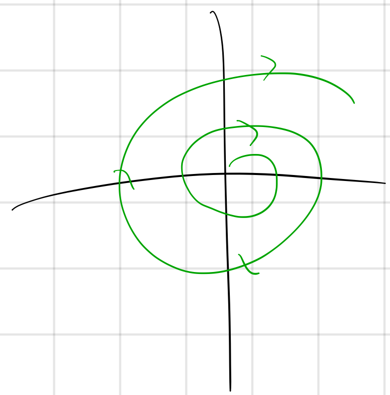
$\lambda = a \pm bi$



stable spiral

$\tau < 0$

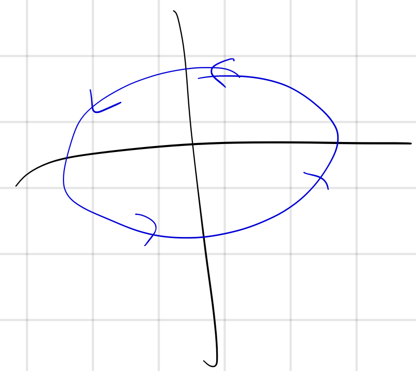
real part negative



unstable spiral

$\tau > 0$

real part positive

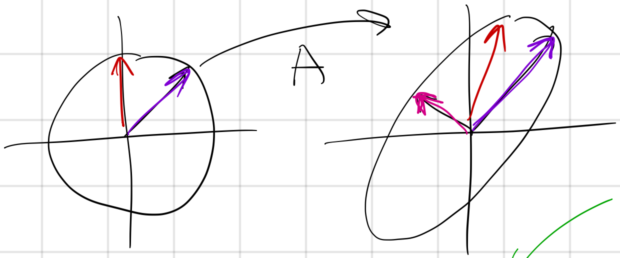


center

$\tau = 0$

real part zero

$$A\vec{v} = \lambda\vec{v}$$



$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Identity matrix

$$\boxed{\det(A - \lambda I) = 0} \Leftarrow \text{characteristic equation}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

$$ad - a\lambda - d\lambda + \lambda^2 - bc = 0$$

$$\lambda^2 - (a+d)\lambda + (ad - bc) = 0$$

$$\boxed{\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0}$$

$$\tau = \operatorname{tr} A = a + d$$

$$\Delta = \det A = ad - bc$$

$$\boxed{\lambda^2 - \tau\lambda + \Delta = 0}$$

$$\boxed{\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}}$$

$$\lambda_1 \lambda_2 = \Delta$$

$$\boxed{\lambda^2 - \tau\lambda + \Delta = 0}$$

$$\boxed{\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}}$$

$\lambda_1 \lambda_2 = \Delta$

$(+)(+) = (+)$

$(-)(-) = (+)$

$(+)(-) = (-)$

$(-)(+) = (-)$

$\lambda_1 + \lambda_2 = \tau$

$(+) + (+) = (+)$

$(-) + (-) = (-)$

To classify fixed point (critical point)

- Δ
  - $(-)$: saddle node (saddle point)
  - (0): inconclusive
  - $(+)$: $\tau^2 - 4\Delta$
    - $(-)$ imaginary: τ
      - $(-)$: stable spiral
      - (0): center
      - $(+)$: unstable spiral
    - $(+)$ real: τ
      - $(-)$: stable node (local max)
      - $(+)$: unstable node (local min)

$$A = \begin{bmatrix} a & 0 \\ c & d \end{bmatrix}$$

lower triangular

$$A = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$$

upper triangular

$$\lambda = a, d$$

$$\vec{x}(t) = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$$